

Project Title:

Airline Data Management and Analysis Using Power BI

Problem Statement:

The airline industry operates with numerous complexities, requiring effective data management and insights into flight schedules, passenger details, and ticketing systems. This project aims to analyze airline operations for improving efficiency and customer satisfaction.

Datasets Used:

Flight\_Information Ticket\_Information Passenger\_Information

1. Flight Information: Includes FlightID, FlightNumber, Airline, Destination, and Status.
2. Passenger Information: Includes PassengerID, FlightID, and SeatNumber.
3. Ticket Information: Includes TicketID, FlightID, and BookingStatus.

Objective:

To analyze and visualize airline data for operational insights, passenger management, and ticket booking trends using Power BI.

Tasks and Marks Distribution

1. Data Preparation and Cleaning (10 Marks)

- Extract and transform data in Power Query.
- Clean data: remove duplicates, handle missing values, and format columns.
- Deliverables: Screenshot of Power Query Editor showing cleaned data.

my last proj

File Home Transform Add Column View Tools Help

Column From Custom Invoke Custom Examples Column Function General

Conditional Column Index Column Duplicate Column

Format Merge Columns Extract Parse

Statistics Standard Scientific Information

From Text From Number From Date & Time

Date Time Duration Text Analytics Vision Azure Machine Learning

Trigonometry Rounding

AI Insights

Queries [3]

flight\_information

Table 1 (ticket\_informati...

passenger\_information

fx = Table.TransformColumnTypes(\*Promoted Headers\*,{{"FlightID", Int64.Type}, {"FlightNumber", type text}, {"Airline", type text}},

Query Settings

PROPERTIES

Name

flight\_information

All Properties

APPLIED STEPS

Source

Navigation

FilterNullAndWhitespace

Removed Other Columns

Promoted Headers

Changed Type

Removed Duplicates

Added Conditional Column

Inserted Text After Delimiter

| FlightID | FlightNumber | Airline | Destination | Status      |           |
|----------|--------------|---------|-------------|-------------|-----------|
| 1        | 1001         | FL1102  | Airline D   | Houston     | On Time   |
| 2        | 1002         | FL1435  | Airline B   | Chicago     | On Time   |
| 3        | 1003         | FL1860  | Airline A   | New York    | Cancelled |
| 4        | 1004         | FL1270  | Airline C   | Chicago     | Delayed   |
| 5        | 1005         | FL1106  | Airline C   | New York    | Delayed   |
| 6        | 1006         | FL1071  | Airline A   | Phoenix     | On Time   |
| 7        | 1007         | FL1700  | Airline C   | Los Angeles | Cancelled |
| 8        | 1008         | FL1020  | Airline C   | Los Angeles | Delayed   |
| 9        | 1009         | FL1614  | Airline A   | Los Angeles | Cancelled |
| 10       | 1010         | FL1121  | Airline D   | Chicago     | Cancelled |
| 11       | 1011         | FL1466  | Airline A   | Phoenix     | On Time   |
| 12       | 1012         | FL1214  | Airline D   | New York    | Delayed   |
| 13       | 1013         | FL1330  | Airline C   | Houston     | On Time   |
| 14       | 1014         | FL1458  | Airline C   | New York    | Delayed   |
| 15       | 1015         | FL1087  | Airline C   | Houston     | Delayed   |
| 16       | 1016         | FL1372  | Airline B   | New York    | Delayed   |
| 17       | 1017         | FL1099  | Airline D   | Phoenix     | Delayed   |
| 18       | 1018         | FL1871  | Airline B   | Houston     | Delayed   |
| 19       | 1019         | FL1663  | Airline B   | Chicago     | Cancelled |
| 20       | 1020         | FL1130  | Airline A   | New York    | On Time   |
| 21       | 1021         | FL1661  | Airline B   | New York    | Cancelled |
| 22       | 1022         | FL1308  | Airline A   | Houston     | Delayed   |
| 23       | 1023         | FL1769  | Airline A   | Chicago     | On Time   |
| 24       | 1024         | FL1343  | Airline B   | Chicago     | Delayed   |
| 25       | 1025         | FL1491  | Airline D   | Phoenix     | On Time   |
| 26       | 1026         | FL1413  | Airline D   | Chicago     | Cancelled |
| 27       | 1027         | FL1805  | Airline D   | Chicago     | On Time   |
| 28       | 1028         | FL1385  | Airline D   | Chicago     | On Time   |

5 COLUMNS, 200 ROWS Column profiling based on top 1000 rows

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File Home Transform Add Column View Tools Help

Column From Custom Invoke Custom Examples Column Function General

Conditional Column Index Column Duplicate Column

Format Merge Columns Extract Parse

Statistics Standard Scientific Information

From Text From Number From Date & Time

Date Time Duration Text Analytics Vision Azure Machine Learning

Trigonometry Rounding

AI Insights

Queries [3]

flight\_information

Table 1 (ticket\_informati...

passenger\_information

fx = Table.Distinct(\*"Changed Type", {"TicketID"})

Query Settings

PROPERTIES

Name

Table 1 (ticket\_informati...

All Properties

APPLIED STEPS

Source

Navigation

FilterNullAndWhitespace

Removed Other Columns

Promoted Headers

Changed Type

Removed Duplicates

| TicketID | FlightID | BookingStatus  |
|----------|----------|----------------|
| 1        | S001     | 1178 Pending   |
| 2        | S002     | 1078 Confirmed |
| 3        | S003     | 1117 Cancelled |
| 4        | S004     | 1120 Cancelled |
| 5        | S005     | 1137 Cancelled |
| 6        | S006     | 1162 Pending   |
| 7        | S007     | 1076 Pending   |
| 8        | S008     | 1035 Cancelled |
| 9        | S009     | 1001 Cancelled |
| 10       | S010     | 1040 Cancelled |
| 11       | S011     | 1064 Pending   |
| 12       | S012     | 1150 Cancelled |
| 13       | S013     | 1060 Cancelled |
| 14       | S014     | 1064 Confirmed |
| 15       | S015     | 1093 Confirmed |
| 16       | S016     | 1072 Pending   |
| 17       | S017     | 1011 Cancelled |
| 18       | S018     | 1105 Cancelled |
| 19       | S019     | 1014 Confirmed |
| 20       | S020     | 1060 Pending   |
| 21       | S021     | 1030 Confirmed |
| 22       | S022     | 1035 Confirmed |
| 23       | S023     | 1165 Confirmed |
| 24       | S024     | 1005 Confirmed |
| 25       | S025     | 1083 Cancelled |
| 26       | S026     | 1123 Cancelled |
| 27       | S027     | 1078 Confirmed |
| 28       | S028     | 1154 Pending   |

3 COLUMNS, 50 ROWS Column profiling based on top 1000 rows

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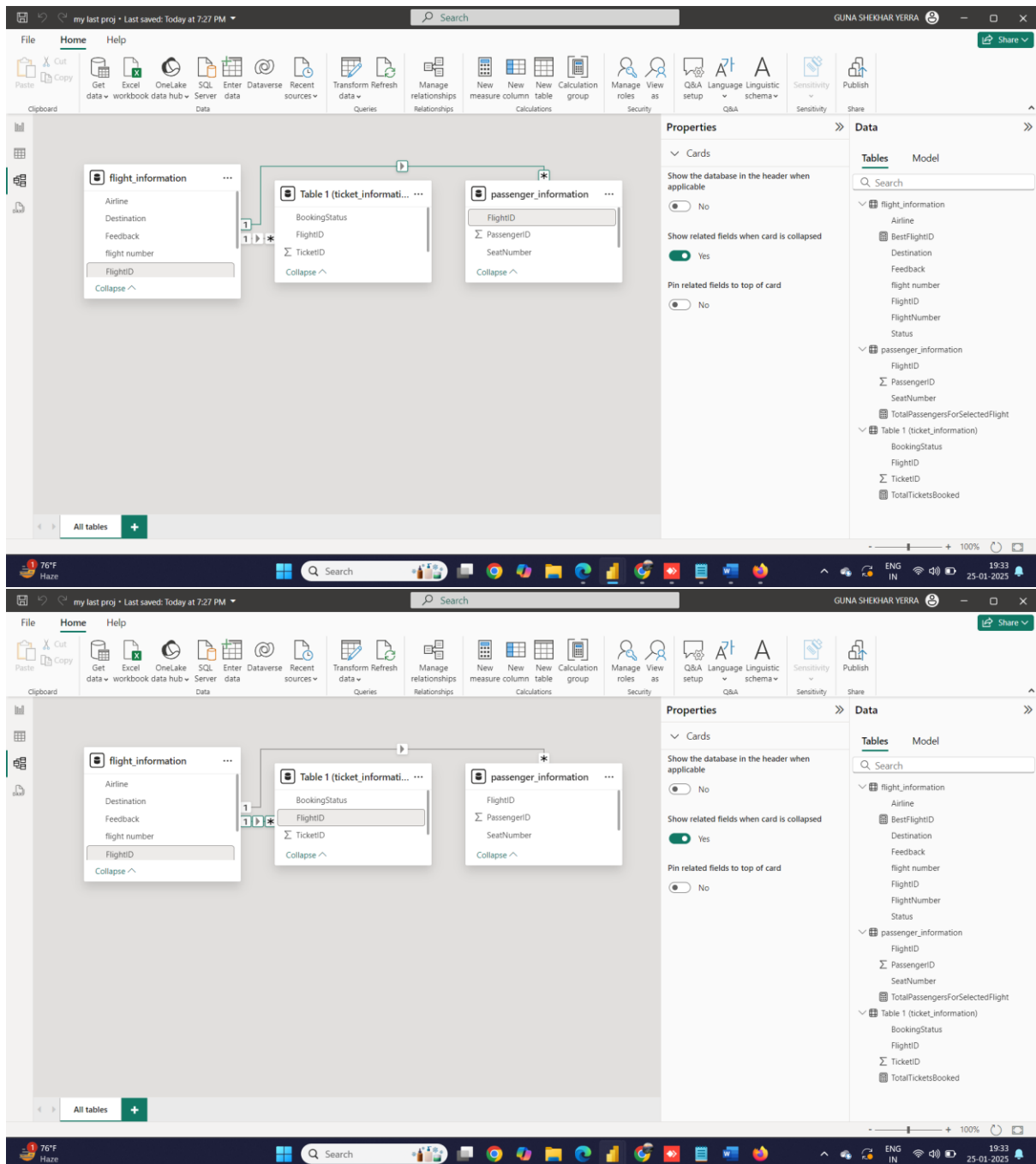
The screenshot displays the Power BI Desktop environment. The main view shows a table with the following data:

| PassengerID | FlightID | SeatNumber |
|-------------|----------|------------|
| 1           | 1        | 1161 38A   |
| 2           | 2        | 1157 24D   |
| 3           | 3        | 1141 30B   |
| 4           | 4        | 1046 17E   |
| 5           | 5        | 1035 29D   |
| 6           | 6        | 1134 10A   |
| 7           | 7        | 1082 10A   |
| 8           | 8        | 1115 20E   |
| 9           | 9        | 1197 34E   |
| 10          | 10       | 1047 2E    |
| 11          | 11       | 1153 43C   |
| 12          | 12       | 1194 48C   |
| 13          | 13       | 1010 47A   |
| 14          | 14       | 1056 23C   |
| 15          | 15       | 1030 16D   |
| 16          | 16       | 1109 40D   |
| 17          | 17       | 1005 25C   |
| 18          | 18       | 1119 32C   |
| 19          | 19       | 1033 27E   |
| 20          | 20       | 1118 32B   |
| 21          | 21       | 1065 19E   |
| 22          | 22       | 1146 5B    |
| 23          | 23       | 1177 28B   |
| 24          | 24       | 1011 22E   |
| 25          | 25       | 1085 6A    |
| 26          | 26       | 1026 5A    |
| 27          | 27       | 1063 12B   |
| 28          | 28       | 1086 46B   |

The right sidebar shows the 'Query Settings' pane. Under 'APPLIED STEPS', 'Removed Duplicates1' is selected. The 'PROPERTIES' section shows the name 'passenger\_information'.

## 2. Data Modeling (10 Marks)

- Create relationships between datasets (FlightID as the key).
- Understand cardinality and configure the model appropriately.
- Deliverables: Screenshot of the data model with relationships.



### 3. Enhanced Data Insights (10 Marks)

- Add a conditional column to classify flights as "Best" or "To Be Improved" based on status.
- Use "Column from Examples" to extract the flight number from FlightNumber.
- Deliverables: Screenshot of the transformed data.

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File Home Transform Add Column View Tools Help

Column From Custom Invoke Custom Examples Column Function Duplicate Column

Conditional Column Index Column Merge Columns Extract Parse From Text From Number From Date & Time

Statistics Standard Scientific Information Date Time Duration Text Analytics Vision Azure Machine Learning

Queries [3]

flight\_information

Table 1 (ticket\_informati...

passenger\_information

Table.AddColumn(#"Added Conditional Column", "flight number", each Text.AfterDelimiter([FlightNumber], "L"), type text)

Query Settings

PROPERTIES

Name

flight\_information

APPLIED STEPS

Source

Navigation

FilterNullAndWhitespace

Removed Other Columns

Promoted Headers

Changed Type

Removed Duplicates

Added Conditional Column

Inserted Text After Delimiter

7 COLUMNS, 200 ROWS Column profiling based on top 1000 rows

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#### 4. Calculations Using DAX (10 Marks)

- Calculate:
- Total passengers for a specific flight.

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Search

GUINA SHEKHAR YERRA

File Home Insert Modeling View Optimize Help

Paste Copy Get Excel OneLake SQL Enter data Datasense Recent sources Transform Refresh New visual Text box More visuals New visual calculation measure measure Sensitivity Publish Copilot

Clipboard

FlightID TotalPassengersForSelectedFlight

TotalPassengersForSelectedFlight by FlightID

Filters

Visualizations

Data

Page 1 of 8

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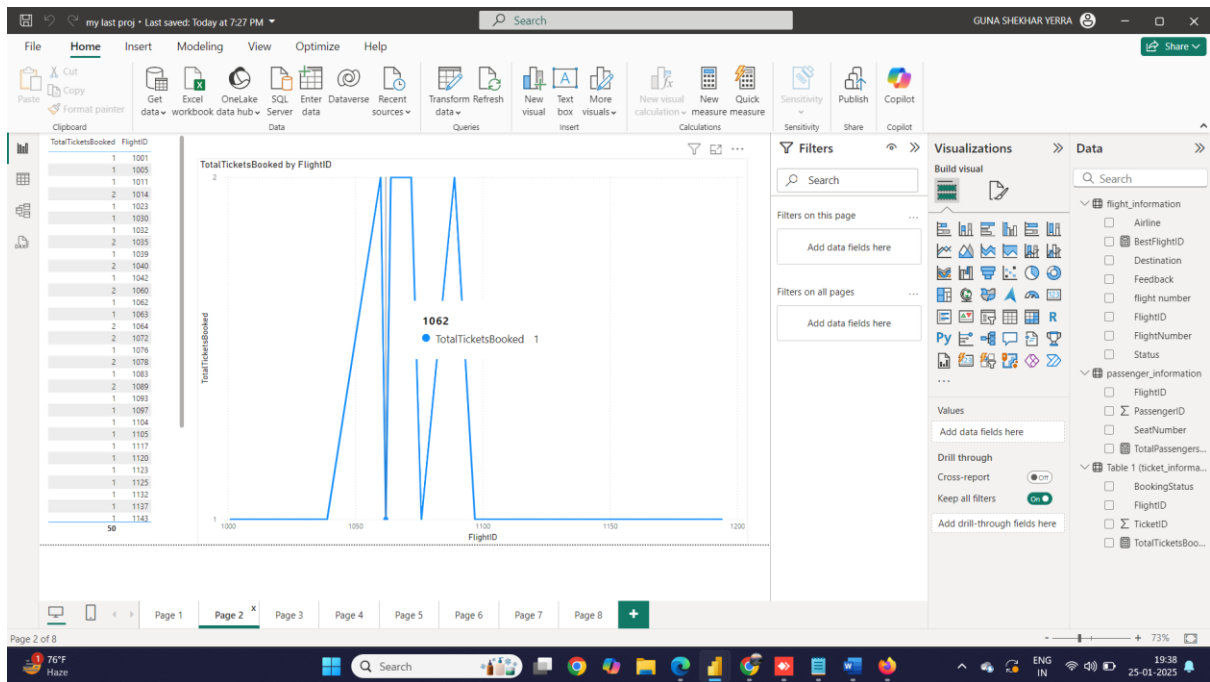
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- Total tickets booked.



o Filtered table showing "Best" flights only.

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GUNA SHEKHAR YERRA

File Home Insert Modeling View Optimize Help

Clipboard: Cut, Copy, Paste, Format painter

Get data: Excel, OneLake, SQL, Enter data, Datasource, Recent sources

Transform Refresh data: Queries

New visual: Text box, More visuals, Insert

New visual calculation: New measure, Quick measure, Measure

Sensitivity: Sensitivity, Publish, Copilot

Visualizations: Build visual

Filters: Search, Filters on this page, Add data fields here, Filters on all pages, Add data fields here

Data: Search, flight\_information (Airline, BestFlightID, Destination, Feedback, flight number, FlightID, FlightNumber, Status), passenger\_information (FlightID, PassengerID, SeatNumber, TotalPassengers...), Table 1 (ticket\_informa... (BookingStatus, FlightID, TicketID, TotalTicketsBoo...)

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Search

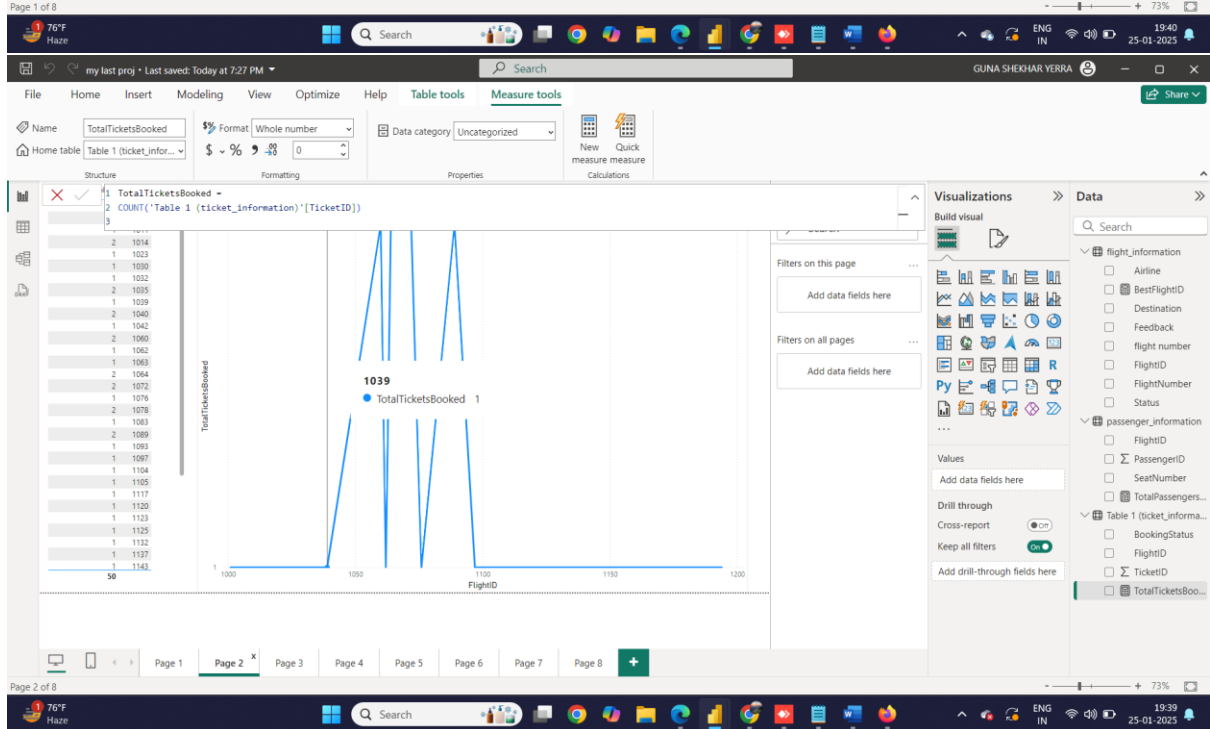
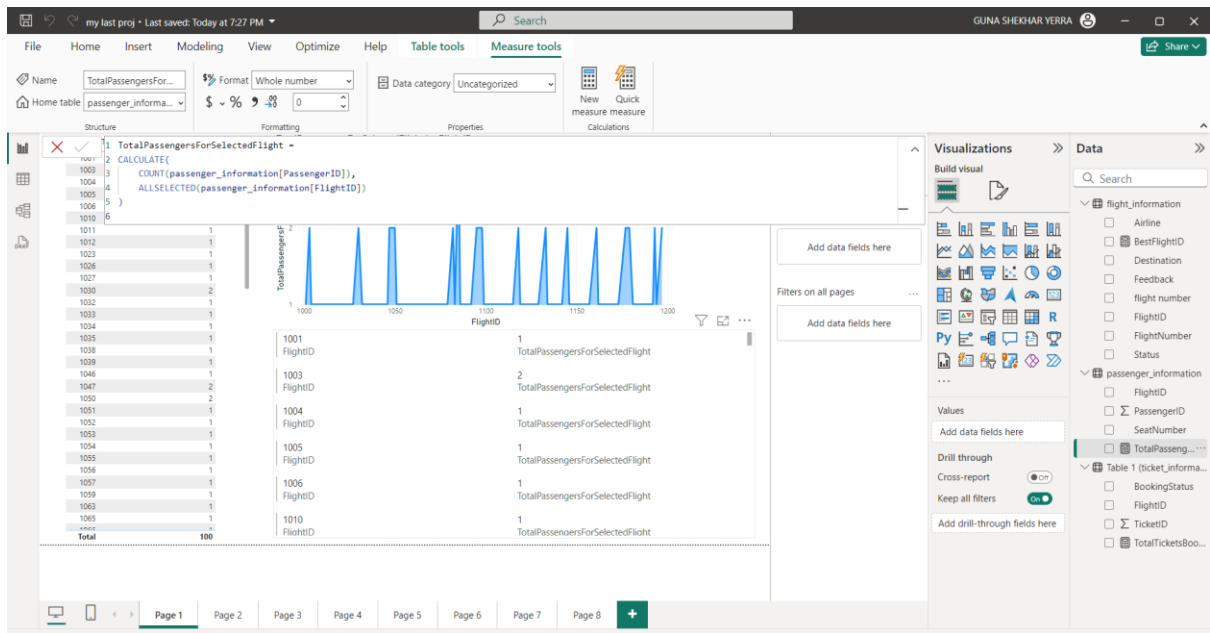
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| BestFlightID | FlightID | Status  | Feedback |
|--------------|----------|---------|----------|
| 1001         | 1001     | On Time | Best     |
| 1002         | 1002     | On Time | Best     |
| 1006         | 1006     | On Time | Best     |
| 1011         | 1011     | On Time | Best     |
| 1013         | 1013     | On Time | Best     |
| 1020         | 1020     | On Time | Best     |
| 1023         | 1023     | On Time | Best     |
| 1025         | 1025     | On Time | Best     |
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| 1046         | 1046     | On Time | Best     |
| 1048         | 1048     | On Time | Best     |
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| 1052         | 1052     | On Time | Best     |
| 1053         | 1053     | On Time | Best     |
| 1055         | 1055     | On Time | Best     |
| 1057         | 1057     | On Time | Best     |
| 1060         | 1060     | On Time | Best     |
| 1061         | 1061     | On Time | Best     |
| 1062         | 1062     | On Time | Best     |
| 1064         | 1064     | On Time | Best     |
| 1071         | 1071     | On Time | Best     |
| 1200         |          |         |          |

● Deliverables: Screenshot of DAX calculations and results.



The screenshot shows the Microsoft Power BI Desktop interface. The Formula Bar contains the following DAX formula:

```

1 BestFlightID =
2 MAXX(
3     FILTER(
4         'flight_information',
5         'flight_information'[Feedback] = "Best"
6     ),
7     'flight_information'[FlightID]
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```

Below the formula, a data table is displayed with columns: FlightID, On Time, and Best. The table contains 100 rows of data, with the last row being 1200, On Time, Best.

The right-hand pane shows the Visualizations pane with a bar chart selected. The Data pane shows the 'flight\_information' table with columns: Airline, BestFlightID, Destination, Feedback, flight number, FlightID, FlightNumber, Status, passenger\_information, FlightID, PassengerID, SeatNumber, TotalPassengers, Table 1 (ticket\_informa..., BookingStatus, FlightID, TicketID, TotalTicketsBoo...

## 5. Visualization and Interactive Features (20 Marks)

- Create visuals for:
- Passenger count by airline.

The screenshot shows the Microsoft Power BI Desktop interface. The main visualization is a bar chart titled "TotalTicketsBooked and Count of TicketID by BookingStatus". The chart displays the count of TicketID for three booking statuses: Cancelled, Confirmed, and Pending. The Y-axis is labeled "TotalTicketsBooked and Count of TicketID" and ranges from 0 to 20. The X-axis is labeled "BookingStatus".

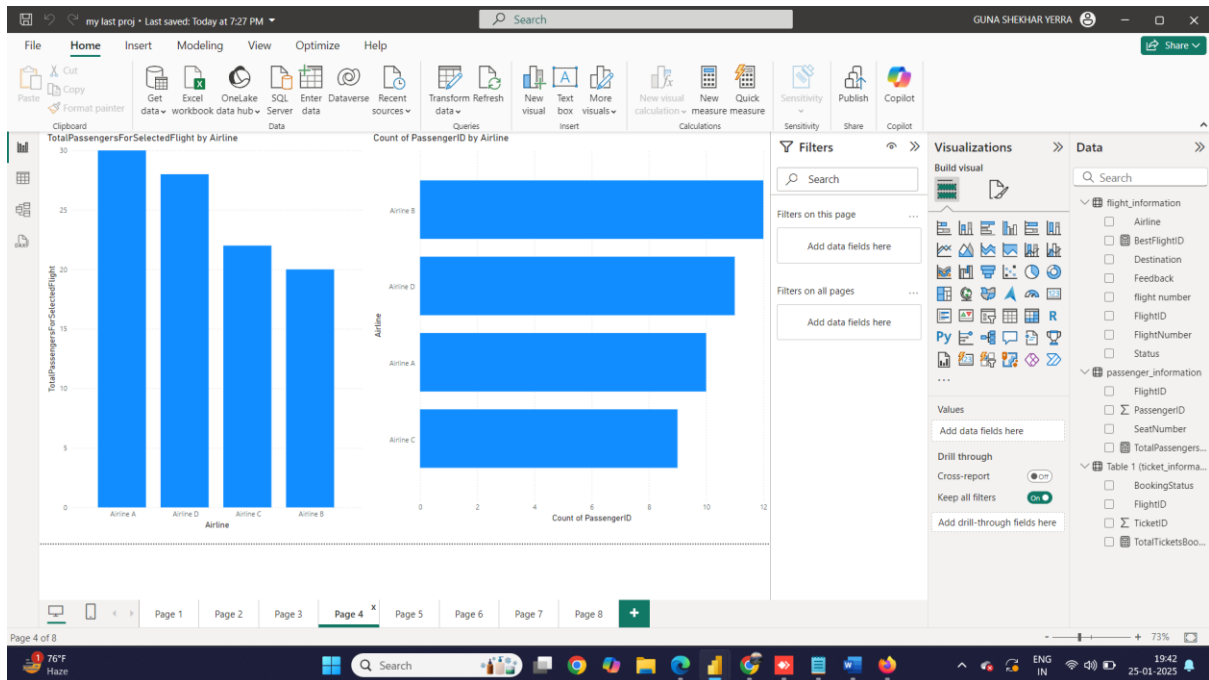
The chart shows the following data:

| BookingStatus | Count of TicketID |
|---------------|-------------------|
| Cancelled     | 18                |
| Confirmed     | 15                |
| Pending       | 12                |

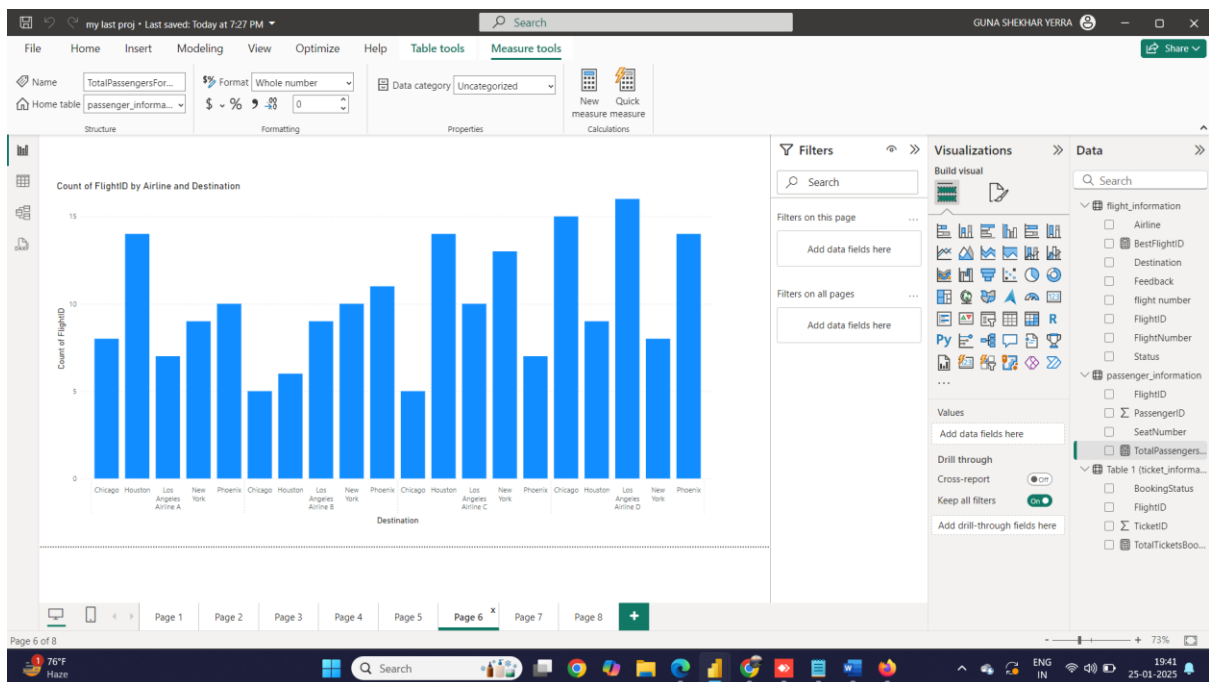
The right-hand pane shows the Visualizations pane with a bar chart selected. The Data pane shows the 'flight\_information' table with columns: Airline, BestFlightID, Destination, Feedback, flight number, FlightID, FlightNumber, Status, passenger\_information, FlightID, PassengerID, SeatNumber, TotalPassengers, Table 1 (ticket\_informa..., BookingStatus, FlightID, TicketID, TotalTicketsBoo...



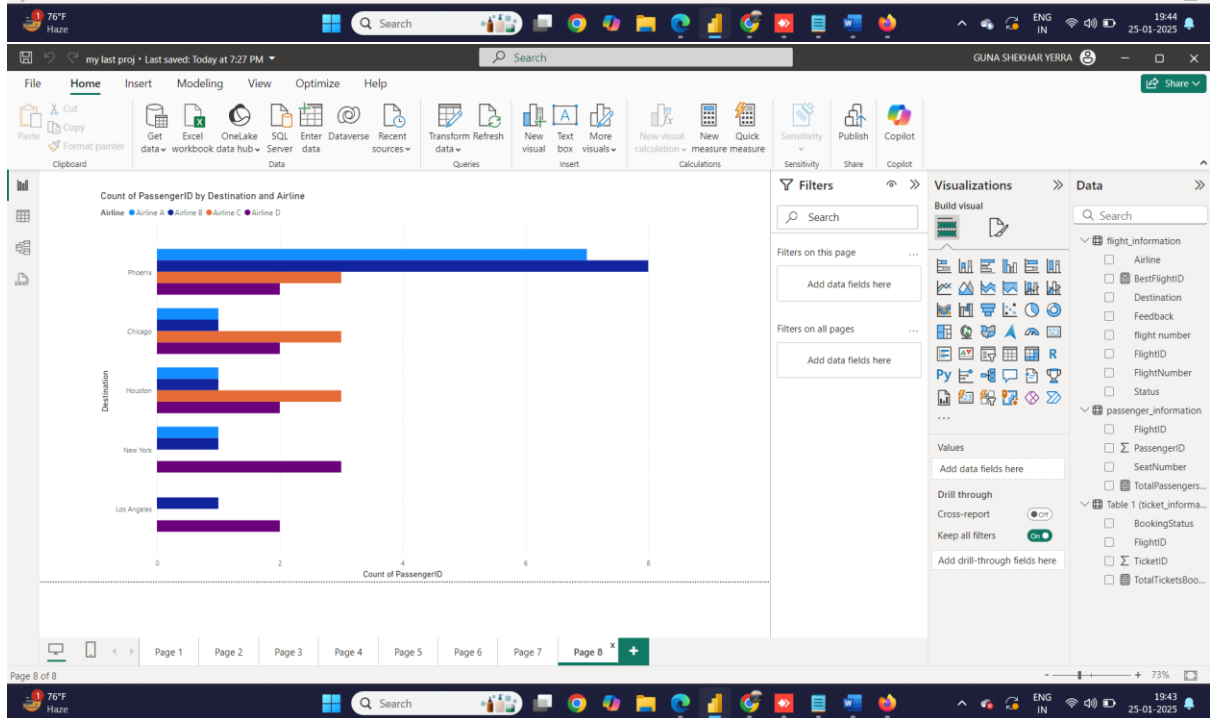
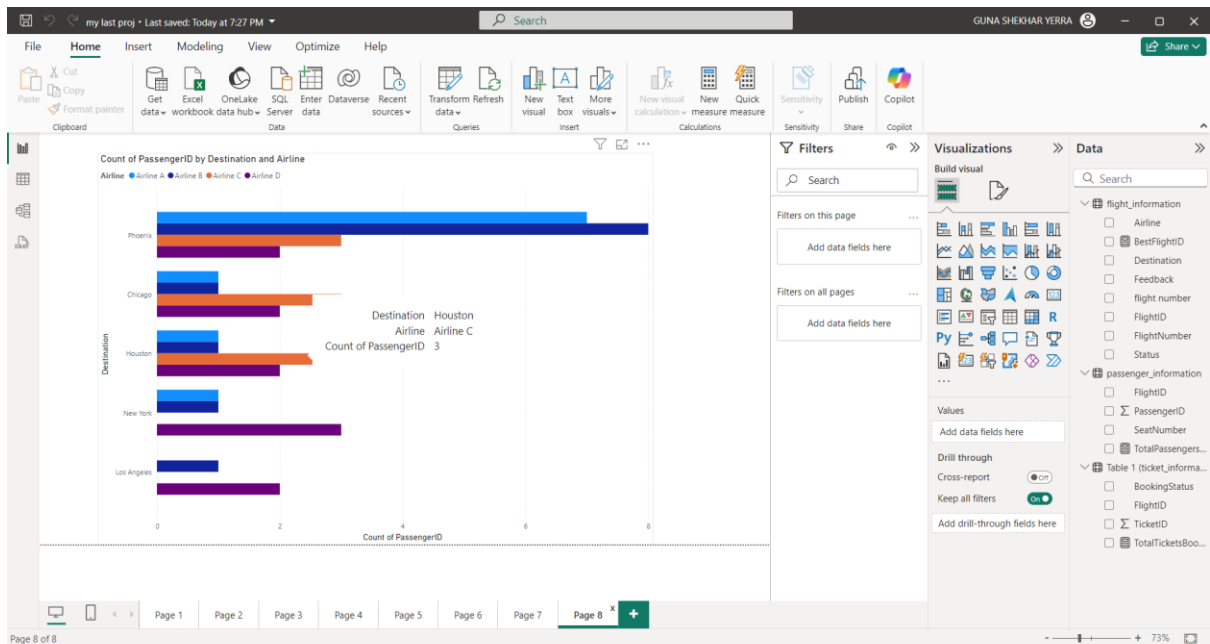
- o Ticket booking statuses.

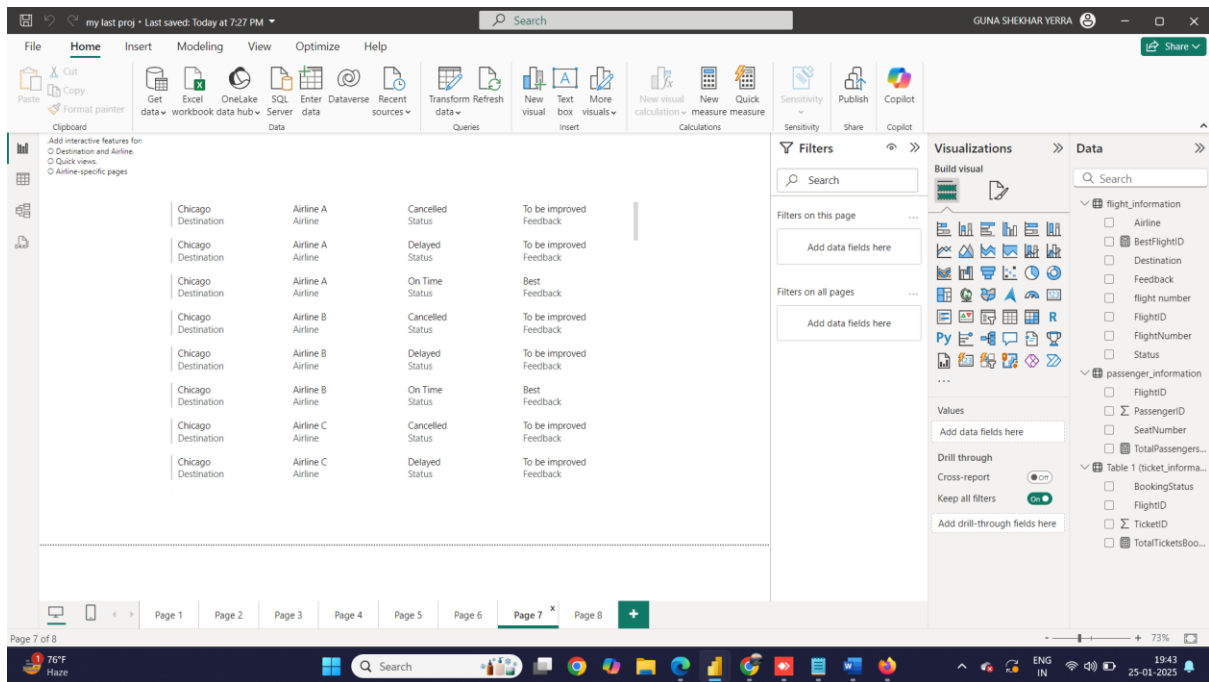


- o Flights by airline and destination.



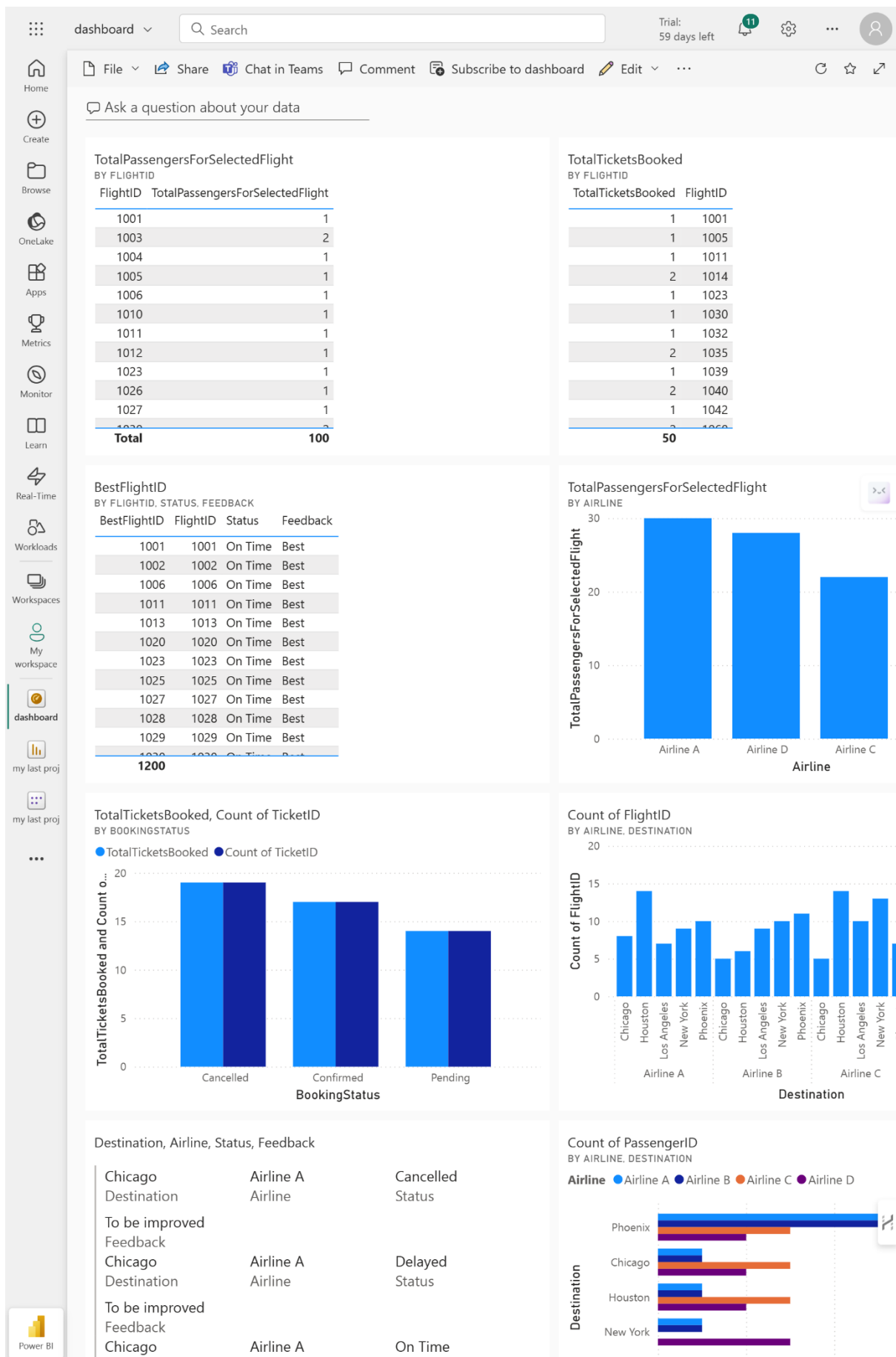
- Add interactive features for:
  - o Destination and Airline.
  - o Quick views.
  - o Airline-specific pages.
- Deliverables: Screenshots of all visuals and interactive features.





## 6. Final Dashboard and Power BI Service (20 Marks)

- Design a comprehensive dashboard with key visuals and insights.



- Configure Row-Level Security (RLS) for Airline A data and assign it to a user.

The top screenshot shows the 'Manage security roles' dialog box in Power BI Desktop. The 'Roles' section lists a new role named 'chickago'. The 'Select tables' section shows 'flight\_informat...', 'passenger\_inf...', and 'Table 1 (ticket\_informa...'. The 'Filter data' section shows a rule: 'Airline' equals 'Airline A'. The 'Filter data' section also includes a 'Switch to DAX editor' button and a 'Show data if' dropdown set to 'All of these rules are true'.

The bottom screenshot shows the main Power BI interface. The 'Visualizations' pane on the right shows a bar chart for 'TotalPassengersForSelectedFlight' and a table for 'Table 1 (ticket\_informa...'. The 'Filters on this visual' pane shows 'Airline is (All)' and 'Count of PassengerID is (All)'. The 'Filters on this page' pane shows 'Airline' and 'Count of PassengerID'. The 'Filters on all pages' pane shows 'Airline' and 'Count of PassengerID'. The 'Visualizations' pane shows a bar chart for 'TotalPassengersForSelectedFlight' and a table for 'Table 1 (ticket\_informa...'. The 'Filters on this visual' pane shows 'Airline is (All)' and 'Count of PassengerID is (All)'. The 'Filters on this page' pane shows 'Airline' and 'Count of PassengerID'. The 'Filters on all pages' pane shows 'Airline' and 'Count of PassengerID'.

- Set up a schedule refresh at 5 PM daily.

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app.powerbi.com/groups/me/settings/datasets/a44f9456-4259-4c80-85a0-e4bf0bc5c35e?experience=power-bi

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Power BI My workspace

Search

Trial: 59 days left

Home Create Browse OneLake Workspaces My workspace dashboard my last proj my last proj

Refresh

Time zone

Time zone configuration is applied not only to determine the schedule refresh time but also to establish the current date and time for incremental refresh models during on-demand and API refreshes.[Learn more](#)

(UTC) Coordinated Universal Time

Configure a refresh schedule

Define a data refresh schedule to import data from the data source into the semantic model. [Learn more](#)

On

Refresh frequency

Daily

Time

5 00 PM

[Add another time](#)

Send refresh failure notifications to

☒ Semantic model owner

☐ These contacts:

Enter email addresses

Apply Discard

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